

2022

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : CC-12P

(OOps Lab using Java)

Full Marks : 30

Batch - 1

Answer *any one* question.

1. Write a Java method to count all words in a string and reverse every word and display them.
2. Write a program in Java to create your own exception as Negative Exception whenever negative values are put in an array.
3. A class called Mypoint, which models a 2D point with x and y co-ordinates. It contains :
 - (a) Two instance variables $x(\text{int})$ and $y(\text{int})$.
 - (b) A default constructor that construct a point at the default location of $(0, 0)$.
 - (c) A overloaded constructor that construct a point with the given x and y co-ordinates.
 - (d) A method `getData()` to take values of x and y from user.
 - (e) A method called `lineSegment(MyPoint  $m$ , MyPoint  $n$ )` that find out the gradient of the line segment and returns it from the function.Write the MyPoint class in Java and also write a class Gradient-check to test all the public methods defined in the class MyPoint.

4. Write a program in Java to create a base class 'Square' having instance variable 'side : double'. Initiate variable using constructor, a method 'getVolume() : double' that calculates volume and print it. Create a derived class 'Cylinder' having instance variable 'height : double'. Initiate variables of both classes through constructor, override method 'getVolume() : double' to calculate volume of cylinder taking 'side' variable of base class as 'radius' and print it.

(2)

X(5th Sm.)-Computer Science-H/Pr./CC-12P/Batch-1

5. Write a Java program to find the longest consecutive elements sequence from a given unsorted array of integers and display it with its length.

Ex : Sample array : [1, 30, 5, 2, 40, 4, 50, 3]. The longest consecutive elements sequence is [1, 2, 3, 4, 5], therefore the program will display [1, 2, 3, 4, 5] and its length 5.

6. Write a Java program to print all permutations of a given string of length 3 (repetition not allowed).
Eg. If the given string is ABC then the permuted strings are ABC, ACB, BAC, CBA, CAB, BCA.
-

2022

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : CC-12P
(OOps Lab using Java)

Full Marks : 30

Batch - 2

Answer *any one* question.

1. Write a program in Java with class Employee and do the following operations on it
 - (a) Create two constructor default and with object as parameter to initialize class variable.
 - (b) Create a function 'Calculate' which calculates the HRA, DA and gross on the basic pay of employee and return all values as an object.
HRA = 12% of basic pay, gross = basic pay + HRA + DA
DA = 3% of basic pay.
 - (c) Display the details of the employees whose gross is more than ₹ 50,000.
2. Create a Java program to create and display a singly linked list. Also write a function to reverse the list.
3. Write a program in Java to delete all consonants from an input string and print the resultant string.
4. Create a class 'Box' having parameterized constructor with an object argument to initialize 'length', 'breadth' and 'height' and also create a method 'Volume()' which returns the volume of the box. Write a Java program to create the class and a test driver main class to test all functions mentioned above.

(2)

5. Using methods `charAt( )` and `lengths( )` of string class, write a Java program to print the frequency of each character in a string.
 6. Write a program to create two threads, one prints 'Hello' and other prints 'Hi'.
-